



Renewables from kites and drones

AWES is a wind harvesting technology which uses a tether and an aircraft instead of wind turbines. <https://digit.in/mar19-04>



Worlds largest bee rediscovered

After nearly four decades, researchers have rediscovered Wallace's giant bee which was believed to be extinct. <https://digit.in/mar19-05>

to data from the satellites. Matching the photos of Steve on the ground allowed the scientists to figure out that three Swarm satellites deployed by the European Space Agency, had travelled right through Steve. The electric field instruments on board the satellites had picked up very clear readings of the event. The gases in Steve were flowing at speeds of 6 kilometers per second, while the winds on either side were a much more tame 10 meters per second. There was also a 3000°C jump in temperatures. The band was about 25 kilometers wide, and at an altitude of 300 kilometers. Roger Haagsmans, mission scientist of Swarm remarked, “It is amazing how a beautiful natural phenomenon, seen by observant citizens, can trig-

To further study the phenomena, scientists used another citizen science platform called Aurorasaurus, partially funded by NASA. The website tracks mentions of the word “Aurora” on Twitter, and churns the data from the social media platform into a region, called the aurora oval, on a world map that predicts where the aurorae will appear. Scientists asked enthusiasts from around the world to submit sightings and photographs of Steve on the Aurorasaurus website, or through its smartphone app. This data was used to further study the phenomenon.

The investigation showed that Steve was related to an atmospheric phenomenon that had been studied by scientists since the 1970s, called Subauroral ion drifts (SAID). The

Steve shows that the lower latitudes of the magnetosphere and the ionosphere need to be studied better, and there may be more mysteries in store in the region. Steve is still not completely understood. Researchers wanted to investigate if Steve was caused by the same stream of particles raining down from the Sun, that give rise to the aurorae. To do this, they analysed a Steve documented in 2008, using images from ground based instruments and data from the NOAA's Polar Orbiting Environmental Satellite 17 (POES-17). The results in this study, published in the journal of the American Geophysical Union called Geophysical Research Letters, shows that Steve is not an aurora at all. The instruments on board the satellite did



A timelapse of Steve

ger scientists' curiosity. The ground network and the electric and magnetic field measurements made by Swarm are great tools that can be used to better understand Steve. This is a nice example of society for science.” The scientists now studying the phenomena acknowledge that it would have remained a little understood phenomenon if the citizen scientists had not brought it to their attention.

Steve was believed to be caused by a similar process as the aurorae, but along different magnetic field lines around the Earth. This is a region where the alignment of the Earth's magnetic and electric fields cause ions and electrons to flow rapidly in an east to west direction.

scientists studying SAID had no idea that there was a visual effect. However, the temperatures and speeds of the gas measured within Steve were higher than typical SAID measurements. As Steve appears in tandem with aurorae in the more northern latitudes, the phenomenon shows some kind of chemical or physical connection between the aurorae and Steve. One of the critical inputs from the citizen scientists was a time lapse of Steve, which showed the structure become more diffuse over time. Briefly, the purple arc was accompanied by green streaks, similar to the shape of a picket fence. The appearance of the green structures suggests turbulence and instability in the ionosphere.

not detect any charged particles raining down in the upper atmosphere, even though the ground based cameras captured Steve. This shows that there is some other mysterious process that causes Steve to appear, and it is not the same process from which the aurorae appear. The conclusion of the study was that Steve is not an aurora at all, and the exact process that leads to its formation remains a mystery. Joe Borovsky, a space physicist from the Space Science Institute in Los Alamos, and not related to the study commented, “This is really interesting because we haven't figured it out and when you get a new problem, it's always exciting. It's like you think you know everything and it turns out you don't.” 